

# Snowpark Container Services (SPCS)

[!IMPORTANT]

- SPCS is available only on certain AWS regions and not available for trial accounts
- All Snowpark Containers are run using a defined compute pool.

## Compute Pool

List of available instance families<sup>1</sup>

- CPU\_X64\_XS
- CPU\_X64\_S
- CPU\_X64\_M
- CPU\_X64\_L
- HIGHMEM\_X64\_S
- HIGHMEM\_X64\_M
- HIGHMEM\_X64\_L
- GPU\_NV\_S
- GPU\_NV\_M
- GPU\_NV\_L

## Create

Create a compute pool named `my_xs_compute_pool` with family `CPU_X64_XS`,

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS
```

Create with `if not exists`,

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --if-not-exists
```

Create with initially suspended (**default**: not suspended initially),

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --init-suspend
```

Create with auto suspend(**default:** 3600 secs) set to 2 mins(120 secs) ,

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --auto-suspend-secs=120
```

Create with minimum nodes(scale down) as 1(**default**) and maximum nodes(scale up) as 3

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --min-nodes=1 --max-nodes=3
```

Create with auto resume on service/job request,

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --auto-resume
```

Create with auto-resume disabled,

    [!NOTE] Auto Resume disabled requires the compute pool to be started manually.

```
snow spcs compute-pool create my_xs_compute_pool \  
  --family CPU_X64_XS --no-auto-resume
```

## List Compute Pool

List all available compute pools for current role,

```
snow spcs compute-pool list
```

List compute pools like my\_xs%

```
snow spcs compute-pool list --like 'my_xs%'
```

## Describe Compute Pool

Get details about a compute pool,

```
snow spcs compute-pool describe my_xs_compute_pool
```

## Status of Compute Pool

To know the current status of a compute pool,

```
snow spcs compute-pool status my_xs_compute_pool
```

## Suspend a Compute Pool

Suspend a compute pool,

```
snow spcs compute-pool suspend my_xs_compute_pool
```

## Resume a Compute Pool

Resume a compute pool,

```
snow spcs compute-pool resume my_xs_compute_pool
```

## Properties on Compute Pool

You can set/unset the following properties on a compute pool after it's created,

| Option              | Description             |
|---------------------|-------------------------|
| --min-nodes         | Minimum Node(s)         |
| --max-nodes         | Maximum Nodes(s)        |
| --auto-resume       | Enable Auto Resume      |
| --no-auto-resume    | Disable Auto Resume     |
| --auto-suspend-secs | Auto Suspend in seconds |
| --comment           | Comment                 |

### Set

Add a comment to the compute pool,

```
snow spcs compute-pool set --comment 'my small compute pool'
my_xs_compute_pool
```

### Unset

Remove the comment from compute pool,

```
snow spcs compute-pool unset --comment my_xs_compute_pool
```

## Delete all services on Compute Pool

Delete all services running on a compute pool

```
snow spcs compute-pool stop-all my_xs_compute_pool
```

## Drop Compute Pool

Drop the compute pool,

```
snow spcs compute-pool drop my_xs_compute_pool
```

# Image Registry

## Login

| [!IMPORTANT] This requires Docker on local system

```
snow spcs image-registry login
```

## Token

Get current user token to access image registry,

```
snow spcs image-registry token
```

## Registry URL

Get image registry URL,

```
snow spcs image-registry url
```

# Image Repository

| [!IMPORTANT]

- A Database and Schema is required to create the Image Repository
- Services can't be created using ACCOUNTADMIN, a custom role is required

The SQL script defines role, grants and warehouse.

As ACCOUNTADMIN run the script to setup required Snowflake resources,

- A Role named cheatsheets\_spcs\_demo\_role to create Snowpark Container Services
- A Database named CHEATSHEETS\_DB where the services will be attached to
- A Schema named DATA\_SCHEMA on DB CHEATSHEETS\_DB to hold the image repository.
- A Warehouse cheatsheets\_spcs\_wh\_s which will be used to run query from services.

Set your Snowflake account user name,

```
export SNOWFLAKE_USER=<your snowflake user>
```

Run the spcs\_setup.sql create the aforementioned Snowflake objects,

```
curl https://raw.githubusercontent.com/Snowflake-Labs/sf-  
cheatsheets/main/samples/spcs_setup.sql |  
snow sql --stdin
```

## Create

Create a image repository named `my_image_repository`,

```
snow spcs image-repository create my_image_repository \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

Create with if not exists,

```
snow spcs image-repository create my_image_repository \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role' \  
  --if-not-exists
```

Replace image repository `my_image_repository`,

```
snow spcs image-repository create my_image_repository \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role' \  
  --replace
```

## List Image Repositories

List all image repositories in the database and schema,

```
snow spcs image-repository list \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## URL

Get URL of the image repository `my_image_repository`,

```
snow spcs image-repository url my_image_repository \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## List Images

Let us push a sample image to repository,

```
IMAGE_REPOSITORY=$(snow spcs image-repository url
my_image_repository \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role')
docker pull --platform=linux/amd64 nginx
docker tag nginx "$IMAGE_REPOSITORY/nginx"
docker push "$IMAGE_REPOSITORY/nginx"
```

List all images in repository my\_image\_repository,

```
snow spcs image-repository list-images my_image_repository \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

## List Image Tags

List all tags for image nginx in repository my\_image\_repository,

[!IMPORTANT] The --image-name should be fully qualified name. Use list-images to get the fully qualified image name

```
snow spcs image-repository list-tags my_image_repository \
  --image-name=/CHEATSHEETS_DB/DATA_SCHEMA/my_image_repository/
nginx \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

## Drop

```
snow spcs image-repository drop my_image_repository \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

## Services

Create a SPCS service specification<sup>2</sup> file,

[!TIP] Tools like **jq** can help extract data from the command output e.g. to get the image name

```
export IMAGE=$(snow spcs image-repository list-images
my_image_repository \
```

```

        --database='CHEATSHEETS_DB' \
        --schema='DATA_SCHEMA' --format json | jq -r '[0].image')

cat <<EOF | tee work/service-spec.yaml
spec:
  containers:
    - name: nginx
      image: $IMAGE
      readinessProbe:
        port: 80
        path: /
  endpoints:
    - name: nginx
      port: 80
      public: true
EOF

```

Create a Service named nginx using compute pool my\_xs\_compute\_pool and specification work/service-spec.yaml,

```

snow spcs service create nginx \
  --compute-pool=my_xs_compute_pool \
  --spec-path=work/service-spec.yaml \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'

```

Create a Service if not exists,

```

snow spcs service create nginx \
  --compute-pool=my_xs_compute_pool \
  --spec-path=work/service-spec.yaml \
  --if-not-exists \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'

```

Create with minimum instances 1 (**default**) and maximum instances to be 3,

```

snow spcs service create nginx \
  --compute-pool=my_xs_compute_pool \
  --spec-path=work/service-spec.yaml \
  --min-instances=1 \
  --max-instances=3 \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'

```

Create service that uses a specific warehouse cheatsheets\_spcs\_wh\_s for all its queries,

```
snow spcs service create nginx \  
  --compute-pool=my_xs_compute_pool \  
  --spec-path=work/service-spec.yaml \  
  --query-warehouse='cheatsheets_spcs_wh_s' \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## Status

Check service status,

[!NOTE] It will take few minutes for the service to be in  
READY status

```
snow spcs service status nginx \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## Describe

Get more details about the service,

```
snow spcs service describe nginx \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## Check Logs of Service

Check the logs of service with the container named nginx with  
instance 0,

[!NOTE] Find instanceId and containerName using the  
command describe command.

```
snow spcs service logs nginx \  
  --container-name=nginx \  
  --instance-id=0 \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

## List

List all available services,

```
snow spcs service list \  
  --database='CHEATSHEETS_DB' \  
  --role='cheatsheets_spcs_demo_role'
```

```
--schema='DATA_SCHEMA' \  
--role='cheatsheets_spcs_demo_role'
```

Query services in database,

```
snow spcs service list --in database CHEATSHEETS_DB \  
--role='cheatsheets_spcs_demo_role'
```

Query services in database and like ng%,

```
snow spcs service list --in database CHEATSHEETS_DB --like 'ng%' \  
\  
--role='cheatsheets_spcs_demo_role'
```

## Service Endpoints

List the service endpoint for the service nginx,

```
snow spcs service list-endpoints nginx \  
--database='CHEATSHEETS_DB' \  
--schema='DATA_SCHEMA' \  
--role='cheatsheets_spcs_demo_role'
```

[!NOTE] Open the ingress\_url on the browser will take you to NGINX home page after authentication

## Suspend a service

Suspend the service,

```
snow spcs service suspend nginx \  
--database='CHEATSHEETS_DB' \  
--schema='DATA_SCHEMA' \  
--role='cheatsheets_spcs_demo_role'
```

## Resume a service

Resume the service,

```
snow spcs service resume nginx \  
--database='CHEATSHEETS_DB' \  
--schema='DATA_SCHEMA' \  
--role='cheatsheets_spcs_demo_role'
```

[!NOTE] Resume service will take few minutes, use the status command to check the status

## Supported properties on Service

You can set/unset the following properties on a service even after it's created,

| Option            | Description                                                              |
|-------------------|--------------------------------------------------------------------------|
| --min-instances   | Minimum number of service instance(s), typically used while scaling down |
| --max-instances   | Maximum number of service instance(s), typically used while scaling up   |
| --auto-resume     | Enable auto resume                                                       |
| --no-auto-resume  | Disable auto resume                                                      |
| --query-warehouse | The Warehouse to use while doing query from the service                  |
| --comment         | Comment for the service                                                  |

### Set

Add a comment to the service,

```
snow spcs service set --comment 'the nginx service' nginx \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

Use service describe to check on the updated property

### Unset

Remove the comment from the service,

```
snow spcs service unset --comment nginx \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

## Upgrade

Upgrade the service nginx with new specification e.g a tag upgrade or probe updates etc.,

```
snow spcs service upgrade nginx \
  --spec-path=work/service-spec_V2.yaml \
  --database='CHEATSHEETS_DB' \
  --schema='DATA_SCHEMA' \
  --role='cheatsheets_spcs_demo_role'
```

## Drop

Drop a service named nginx

```
snow spcs service drop nginx \  
  --database='CHEATSHEETS_DB' \  
  --schema='DATA_SCHEMA' \  
  --role='cheatsheets_spcs_demo_role'
```

[!NOTE] SPCS has compute associated with it, run the **clean up** script to clean the Snowflake resources created as part of this cheatsheet.

```
curl https://raw.githubusercontent.com/Snowflake-Labs/sf-  
cheatsheets/main/samples/spcs_cleanup.sql |  
snow sql --stdin
```

## Quickstarts

- [Snowflake Developers::Quickstart](#)
- [Snowflake Developers::Getting Started With Snowflake CLI](#)
- [Intro to Snowpark Container Services](#)
- [Build a Data App and run it on Snowpark Container Services](#)

## Documentation

- [Snowflake CLI](#)
- [Execute Immediate Jinja Templating](#)
- [Snowpark Container Services](#)

## Tutorials

- [Snowpark Container Services Tutorial](#)

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1. <https://docs.snowflake.com/en/sql-reference/sql/create-compute-pool>
  2. <https://docs.snowflake.com/en/developer-guide/snowpark-container-services/specification-reference>